

Aron Ukbya Bariagabr

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Professional Summary

Java Full Stack Developer with over 6 years of experience designing, developing, and supporting enterprise applications in the healthcare and financial services industries. Experienced in building scalable web applications, microservices, and cloud-based solutions using Java, Spring Boot, React, AWS, and Azure. Contributed to the development of EHR and PMS platforms at CVS Health, improving workflow efficiency by 26% and supporting systems managing over 1.5 million patient records with 99.9% availability. Developed predictive analytics solutions using Deeplearning4j to enhance business decision making and operational efficiency. Strong background in REST APIs, database design, cloud technologies, CI/CD pipelines, and Agile Scrum methodologies. Passionate about delivering secure, high performance, and reliable software solutions that drive business value.

Technical Skills

Programming Languages: Java, JavaScript (ES6+), SQL, Python, C++
Frameworks & Libraries: Spring Boot, Hibernate, React, Angular, Bootstrap, Tailwind CSS
Web & API Development: RESTful APIs, JSON, WebSocket, Responsive Web Design, Swagger/OpenAPI
Databases: Oracle, MySQL, PostgreSQL, SQL Server
Cloud & DevOps: AWS (EC2, S3, IAM, Lambda, CloudWatch), Azure (App Service, Functions, Key Vault), Docker, Kubernetes, Jenkins, Git, GitHub, Prometheus, ELK Stack
Testing & Development Tools: JUnit, Mockito, Postman, IntelliJ IDEA, Eclipse
Machine Learning & AI: Deeplearning4j (DL4J),GitHub Copilot
Domain Expertise: Patient Management Systems (PMS), Electronic Health Records (EHR), HIPAA Compliance, ERP Systems
Soft Skills: Agile/Scrum Methodology, Root Cause Analysis, Documentation, Stakeholder Engagement, Problem-Solving, Communication, Collaboration
Backend & Architecture: Microservices Architecture, Distributed Systems, System Design, REST API Design, Design Patterns
Security: OAuth2, JWT Authentication, Role Based Access Control (RBAC), Secure API Development
DevOps: CI/CD Pipelines, GitHub Actions or Bitbucket Pipelines
Backend Architecture: High Availability Systems, Fault Tolerant Systems

Professional Experience

Full Stack Java Developer, CVS HealthJan 2025 - Present | TX

- Developed EHR and PMS modules using Java, Spring Boot, Hibernate, and React, optimizing workflows for over 12,000 healthcare users, improving system efficiency by 26% across clinical operations.
- Deployed enterprise-scale healthcare systems on Azure App Service and Functions with PostgreSQL and SQL Server, managing 1.5M+ patient records via Azure Key Vault, ensuring 99.9% uptime and full HIPAA-compliant data governance.
- Engineered RESTful APIs and WebSocket integrations for 50+ medical systems, reducing latency by 18%, and documenting endpoints with Swagger and OpenAPI for seamless interoperability and maintenance.
- Applied Deeplearning4j (DL4J) to build predictive models for clinical risk and operational insights, enhancing decision accuracy by 28%, and generating an estimated \$120K operational impact.
- Implemented secure REST APIs using OAuth2, JWT, and role-based access control (RBAC) to meet HIPAA compliance and enterprise security standards.
- Designed and supported Java microservices and distributed systems with Eclipse and Docker, implementing JWT authentication and MySQL data storage to ensure high availability and secure data access for enterprise healthcare platforms, which enhanced system reliability and compliance.
- Monitored microservices and application health using Prometheus and ELK Stack, identifying performance bottlenecks early and reducing system downtime by 22% across distributed healthcare environments.
- Directed Agile/Scrum releases using Jenkins, Docker, and Kubernetes, executing 8+ monthly deployments, improving build reliability by 35%, and collaborating through GitHub for continuous integration and peer code reviews.

Junior Full Stack Developer, DXC TechnologyJun 2021 - Dec 2024 | CA

- Designed backend system components and microservices architecture for cloud-based financial applications using AWS EC2, Lambda, Docker, and CI/CD pipelines, improving scalability, performance, and deployment reliability.
- Optimized JSON-based RESTful APIs for secure payment processing, reducing response times by 24%, while protecting sensitive financial data using AWS IAM roles and permissions aligned with enterprise compliance policies.
- Maintained Oracle and MySQL databases, managing 5M+ transactions, improving reconciliation efficiency by 27%, and lowering operational costs by approximately \$80,000 annually through query optimization and automation.
- Built internal automation tools and API test utilities using Python, streamlining backend validation and reducing manual QA effort by 25% across development sprints.
- Developed C++ utilities for financial data parsing and performance tuning, enhancing backend transaction processing throughput by 18% on AWS-based systems.
- Executed unit and API testing using JUnit, Mockito, and Postman in IntelliJ IDEA, reducing defects across 12 core financial modules by 32%, improving reliability and customer transaction success rates.
- Coordinated with product and operations teams using Agile/Scrum, integrating Git for version control, ensuring deliverables across three functional sprints aligned with project timelines and client expectations.
- Implemented AWS CloudWatch and S3-based monitoring pipelines, tracking application logs and financial data backups to ensure 99.8% data availability and proactive anomaly detection across distributed systems.

Software Engineering Intern, DXC TechnologyAug 2020 – May 2021 | CA

- Supported backend development tasks using Java and Spring Boot, assisting senior engineers in enhancing enterprise web applications, contributing to a 15% improvement in feature delivery speed.
- Helped debug and validate RESTful APIs, ensuring accurate data flow between internal services, reducing API defects by 12% during sprints.
- Worked with SQL databases (Oracle/MySQL) to verify data consistency and support backend operations, improving query accuracy by 10%.
- Assisted in developing UI components using JavaScript (ES6+), Angular, and Bootstrap, focusing on responsive layout improvements that enhanced usability scores by 14%.
- Executed unit tests using JUnit, helping identify defects and improve overall code stability, lowering bug recurrence by 11%.

- Collaborated with team members using Git and GitHub, participating in code reviews and version control processes, increasing merge efficiency by 9%.
- Participated in Agile/Scrum ceremonies, including sprint planning, daily stand-ups, and delivery workflows, which improved sprint completion rates by 12%.
- Assisted with build verification and deployment readiness checks within CI/CD pipelines, supporting smooth and reliable releases with 10% fewer deployment errors.

Software Engineer, Eritrean Telecommunication Services Corporation (EriTel) Sep 2016-Mar 2017 | Asmara, Eritrea

- Collaborated with a three-member team to design and build a business analytics platform from scratch using Spring Boot for the backend and Bootstrap for the UI, enabling the company to consolidate sales and operational data in one system.
- Developed backend components with Java and Spring Boot and created frontend pages with HTML and Bootstrap, delivering core functionality for data entry and reporting.
- Gathered requirements and implemented features by translating client business needs into technical specifications, using GitHub Copilot to accelerate code generation and ensure alignment with stakeholder goals.
- Designed and implemented features that processed and analyzed sales, investment, advertising, and operational metrics using RESTful APIs and JSON, providing users with actionable insights into performance trends.
- Assisted in building analytics functions that identified business trends, helping management spot profitability issues early and adjust strategies accordingly.
- Debugged application issues using Spring Boot and GitHub Copilot, enhanced functionality, and supported testing, resulting in fewer runtime errors and smoother test cycles.
- Participated in design discussions and implemented Spring Boot-based enhancements across the development lifecycle, improving code modularity and maintainability.
- Worked with clients and stakeholders to gather requirements and delivered RESTful API solutions using JavaScript (ES6+) and JSON, enabling faster integration with client systems.

Education

Bachelor of Science in Computer Engineering University of California San Diego (UCSD), San Diego, CA	Jan 2019 - Jun 2025
Bachelor of Science in Computer Science Eritrea Institute of Technology, May-Nefhi, Maekel, Eritrea	Sep 2013 - Jun 2017

Relevant Coursework: Data Structures & Algorithms, Embedded Systems, Operating Systems, Computer Networks, Software Engineering